

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

## Department of Computer Science and Engineering

### ASSESSMENT TOOL 2 – Concept Mapping

|                  |                                                                                                               |               |                                     |
|------------------|---------------------------------------------------------------------------------------------------------------|---------------|-------------------------------------|
| Course Code:     | CSA0309                                                                                                       | Course Title: | Data Structures                     |
| CO Assessed:     | CO1 – Imitation (BL3, BL4)                                                                                    | Assessment:   | Assessment Tool 2 – Concept Mapping |
| Weightage:       | 35% of CIA                                                                                                    | Total Marks:  | 100                                 |
| CO1 Statement:   | Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity. |               |                                     |
| Student Name:    | Koppala Navya                                                                                                 | Reg. No.:     | 192511231                           |
| Section / Batch: | 2 <sup>nd</sup> Year AIDS                                                                                     | Date:         | 08/07/2026                          |

#### Code of Conduct

I R, Rohini Sree (192524471) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Koppala Navya

#### Instructions

- This test contains 5 Concept mapping questions. Total: 100 Marks.
- When a student answer using a concept map, in evaluation the answer should be with following four requirements: Understanding of concepts, Connections between ideas, Logical structure, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

| Section / Topic                                                                   | Focus                                    | Q Nos | Marks     |
|-----------------------------------------------------------------------------------|------------------------------------------|-------|-----------|
| Linked data structure                                                             | Basic data structure                     | 1     | 20        |
| Primitive and Non-Primitive Data Structure - Linear and Non-Linear Data Structure | Classifications of Linear data structure | 2     | 20        |
| Search Data Structure                                                             | Linear Search and Binary Search          | 3     | 20        |
| Recursion, Performance analysis                                                   | Recursion                                | 4     | 20        |
| Tree Data Structures                                                              | Classifications of Tree data structure   | 5     | 20        |
| GRAND TOTAL:                                                                      |                                          |       | 100 Marks |

## Annexure A – Concept Mapping

1. A smart city traffic management system handles data related to vehicle movement, road networks, and signal control. Construct a concept map connecting Primitive Data Structures → Non-Primitive Data Structures → Linear → Non-Linear Structures. Include examples such as arrays, linked lists, trees, and graphs, and illustrate their applications in traffic management. Explain the relationships between these structures and justify their classification based on functionality and efficiency. (20 Marks)
2. A smart healthcare system processes large volumes of patient data such as temperature readings, medical history, and diagnostic details. Construct a concept map linking the progression from Data → Data Object → Data Structure → Abstract Data Type (ADT) → Implementation in C. Identify relevant sub-concepts such as structures, pointers, and operations, and establish meaningful relationships among them. Organize the concept map hierarchically and justify how each level contributes to efficient data organization, processing, and improved system performance. (20 Marks)
3. E-commerce platform manages product storage, searching, and recommendation systems. Construct a concept map linking Linear Data Structures → Sequential Access → Limitations → Non-Linear Data Structures → Hierarchical/Network Relationships → Applications. Analyze the differences between linear and non-linear structures and demonstrate how each supports specific components of the system. Justify which structure is more efficient for handling complex applications. (20 Marks)
4. A video streaming application must balance speed and memory usage. Construct a concept map connecting Time Complexity → Space Complexity → Tradeoff → Algorithm Design → Performance Optimization. Analyze how improving one factor affects the other and justify the optimal balance. (20 Marks)
5. A digital archive stores millions of documents. Construct a concept map linking Searching → Linear Search → Binary Search → Sorted Data → Time Complexity → Performance. Analyze the impact of data organization on search efficiency and justify the best technique. (20 Marks)

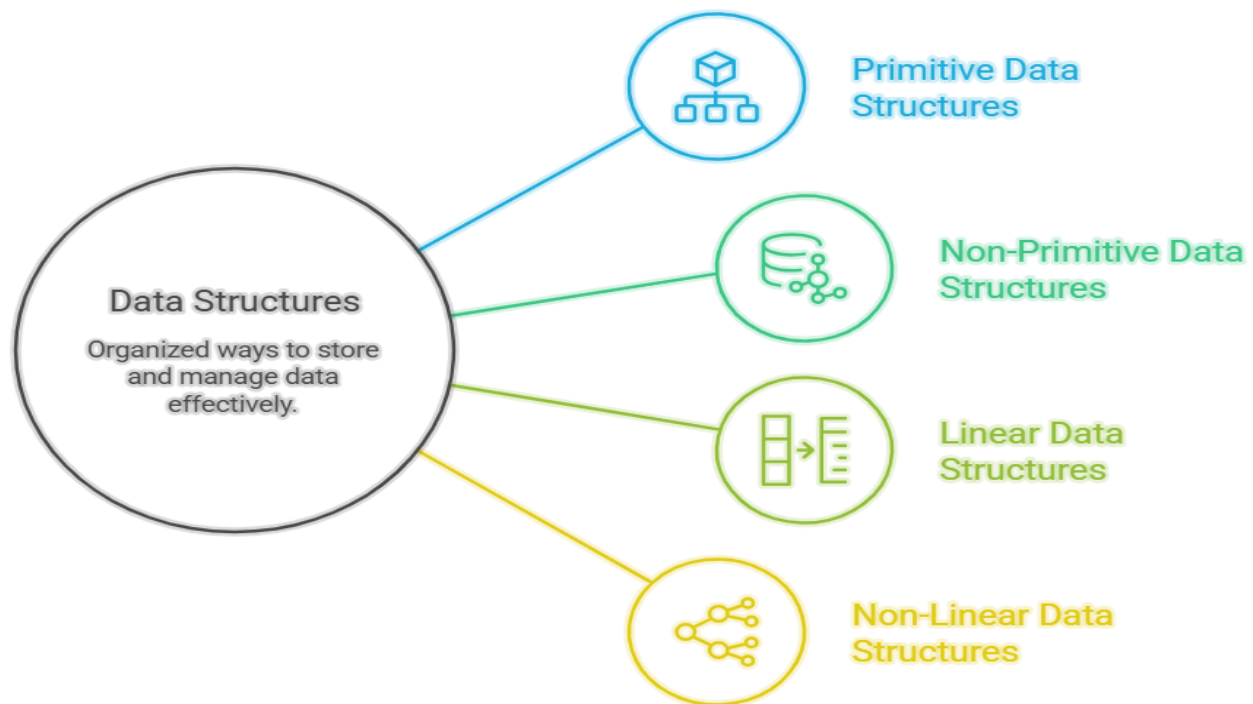
### Rubrics for Calculation

| Criteria                                | Weightage | Level 4 – Exemplary                                                        | Level 3 – Proficient                                  | Level 2 – Developing                           | Level 1 – Beginning                                    |
|-----------------------------------------|-----------|----------------------------------------------------------------------------|-------------------------------------------------------|------------------------------------------------|--------------------------------------------------------|
| <b>Understanding of Problem Context</b> | 20        | Clearly understands the problem and accurately identifies all requirements | Good understanding with minor gaps                    | Basic understanding; some requirements unclear | Poor understanding or misinterpretation of the problem |
| <b>Application of Concepts</b>          | 30        | Accurately applies appropriate concepts to solve complex problems          | Applies relevant concepts correctly with minor errors | Applies basic concepts with limited accuracy   | Fails to apply appropriate concepts                    |
| <b>Problem-Solving Approach</b>         | 20        | Logical, well-structured, and efficient approach                           | Mostly logical approach with minor gaps               | Basic approach with some inconsistencies       | Illogical or unclear approach                          |
| <b>Accuracy of Solution</b>             | 20        | Completely correct solution with proper justification                      | Mostly correct with minor errors                      | Partially correct solution                     | Incorrect or incomplete solution                       |
| <b>Clarity</b>                          | 10        | Clear, well-organized, and properly explained solution                     | Understandable with minor clarity issues              | Limited clarity and explanation                | Poorly presented and difficult to understand           |

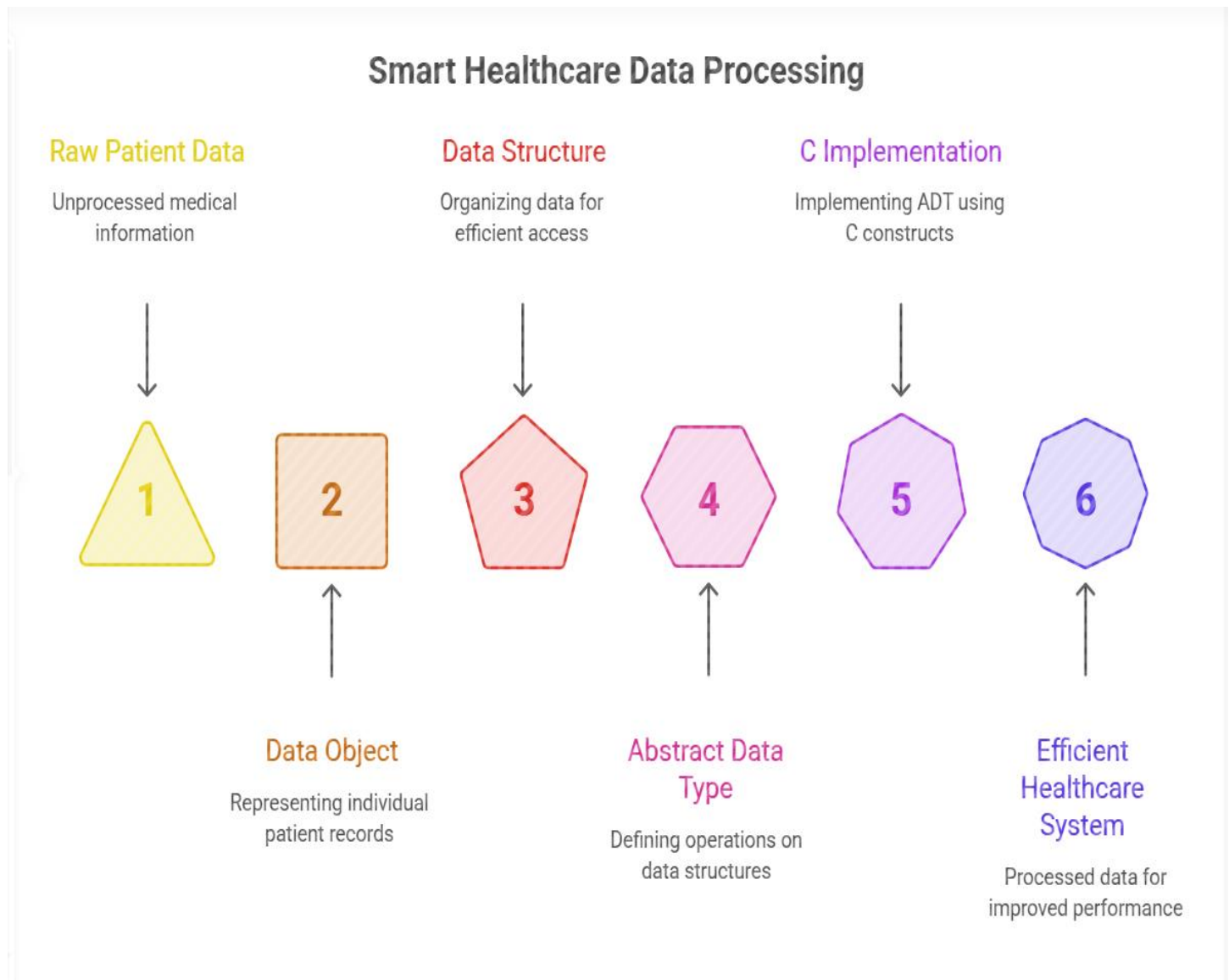
| Faculty In-charge | Course Coordinator | Program Director |
|-------------------|--------------------|------------------|
| Signature: _____  | Signature: _____   | Signature: _____ |
| Date: _____       | Date: _____        | Date: _____      |

## QUESTION-1

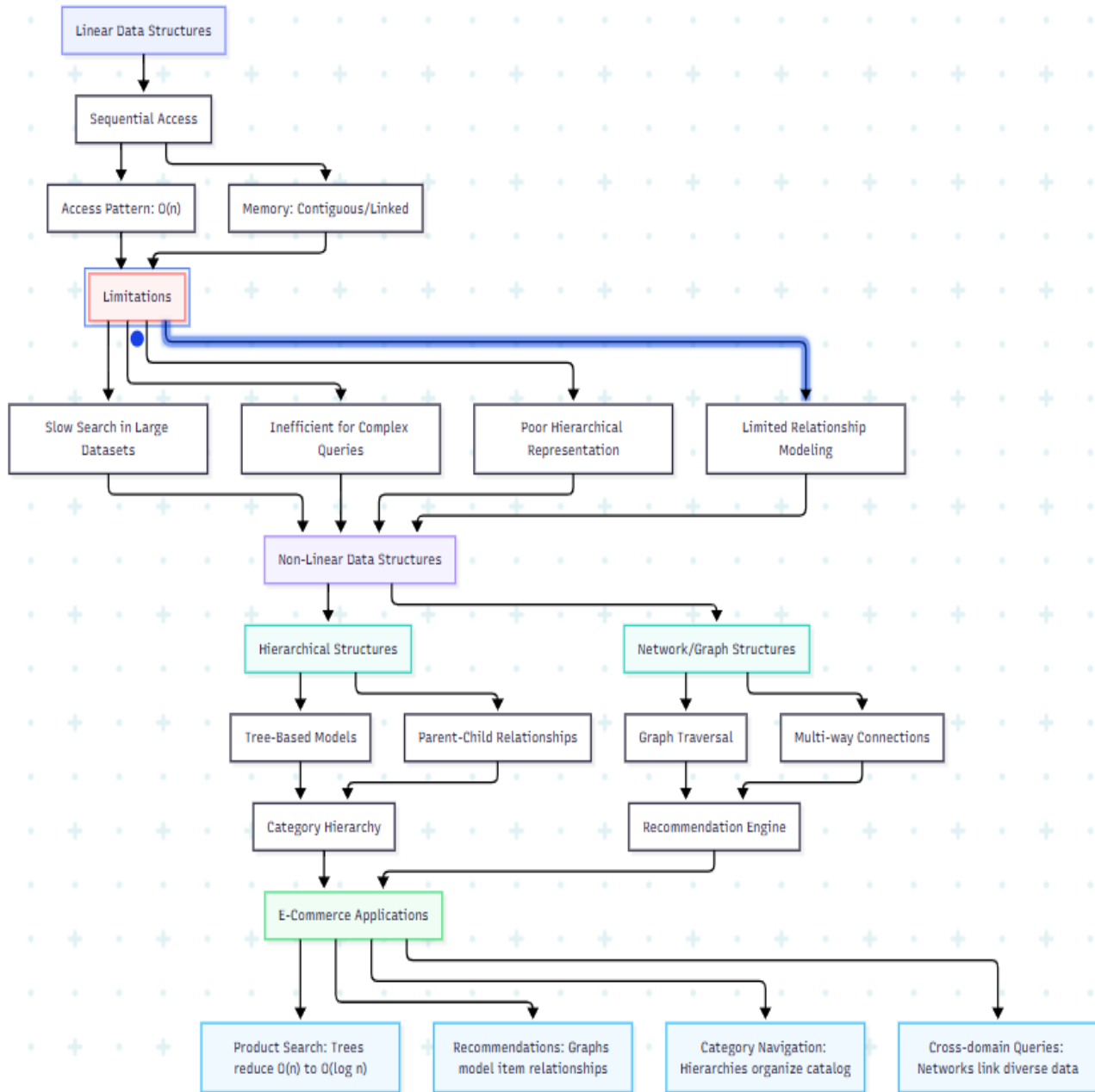
### Unveiling Data Structures in Smart City Traffic Management



## QUESTION-2



### QUESTION-3



## QUESTION-4

### Balancing Speed and Memory in Video Streaming

#### Time Complexity

Measures the execution time of an algorithm

#### Space Complexity

Measures the memory usage of an algorithm

#### Tradeoff

The compromise between time and space

#### Algorithm Design

The process of creating efficient algorithms

#### Performance Optimization

Techniques to improve algorithm efficiency



## QUESTION-5

# Search Techniques in Digital Archives

## Linear Search

Searches through data sequentially, suitable for unsorted data.

## Binary Search

Searches through sorted data efficiently.

## Data Organization

Affects search efficiency based on sorting.

## Time Complexity

Measures the efficiency of search algorithms.

